

1. Student Management System

```
import java.util.Scanner;

public class Student {

    private String name;
    private int rollNo;
    private int marks1, marks2, marks3;

    public void acceptDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student Name: ");
        name = sc.nextLine();

        System.out.print("Enter Roll Number: ");
        rollNo = sc.nextInt();

        System.out.print("Enter Marks 1: ");
        marks1 = sc.nextInt();

        System.out.print("Enter Marks 2: ");
        marks2 = sc.nextInt();

        System.out.print("Enter Marks 3: ");
        marks3 = sc.nextInt();
    }

    public double calculateAverage() {
        return (marks1 + marks2 + marks3) / 3.0;
    }

    public void displayResult() {
        double average = calculateAverage();

        System.out.println("\n----- Student Details -----");
        System.out.println("Name : " + name);
        System.out.println("Roll No : " + rollNo);
        System.out.println("Average Marks : " + average);

        if (average >= 50)
            System.out.println("Result : PASS");
        else
            System.out.println("Result : FAIL");
    }
}
```

```

    }

    public static void main(String[] args) {
        Student s = new Student();
        s.acceptDetails();
        s.displayResult();
    }
}

```

```

1- import java.util.Scanner;
2
3- public class Student {
4
5     private String name;
6     private int rollNo;
7     private int marks1, marks2, marks3;
8
9- public void acceptDetails() {
10     Scanner sc = new Scanner(System.in);
11
12     System.out.print("Enter Student Name: ");
13     name = sc.nextLine();
14
15     System.out.print("Enter Roll Number: ");
16     rollNo = sc.nextInt();
17
18     System.out.print("Enter Marks 1: ");
19     marks1 = sc.nextInt();
20
21     System.out.print("Enter Marks 2: ");
22     marks2 = sc.nextInt();
23
24     System.out.print("Enter Marks 3: ");
25     marks3 = sc.nextInt();
26 }

```

```

Enter Student Name: asif
Enter Roll Number: 192521009
Enter Marks 1: 80
Enter Marks 2: 75
Enter Marks 3: 86

----- Student Details -----
Name : asif
Roll No : 192521009
Average Marks : 80.33333333333333
Result : PASS

=== Code Execution Successful ===

```

2. Library Book Management

```

import java.util.Scanner;

public class LibraryBook {

    // Private data members (Data Hiding)
    private String bookTitle;
    private String author;
    private boolean isIssued;

    // Method to accept book details
    public void acceptDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Book Title: ");
        bookTitle = sc.nextLine();

        System.out.print("Enter Author Name: ");
        author = sc.nextLine();

        isIssued = false; // Initially available
    }

    // Method to issue the book

```

```

public void issueBook() {
    if (!isIssued) {
        isIssued = true;
        System.out.println("Book Issued Successfully.");
    } else {
        System.out.println("Book is already issued.");
    }
}

// Method to return the book
public void returnBook() {
    if (isIssued) {
        isIssued = false;
        System.out.println("Book Returned Successfully.");
    } else {
        System.out.println("Book is already available.");
    }
}

// Method to display book details
public void displayDetails() {
    System.out.println("\n----- Book Details -----");
    System.out.println("Book Title : " + bookTitle);
    System.out.println("Author    : " + author);

    if (isIssued)
        System.out.println("Status    : Issued");
    else
        System.out.println("Status    : Available");
}

// Main Method
public static void main(String[] args) {
    LibraryBook b = new LibraryBook();

    b.acceptDetails();
    b.displayDetails();

    b.issueBook();
    b.displayDetails();

    b.returnBook();
    b.displayDetails();
}
}

```

```

1- import java.util.Scanner;
2
3- public class LibraryBook {
4
5     // Private data members (Data Hiding)
6     private String bookTitle;
7     private String author;
8     private boolean isIssued;
9
10    // Method to accept book details
11- public void acceptDetails() {
12        Scanner sc = new Scanner(System.in);
13
14        System.out.print("Enter Book Title: ");
15        bookTitle = sc.nextLine();
16
17        System.out.print("Enter Author Name: ");
18        author = sc.nextLine();
19
20        isIssued = false; // Initially available
21    }
22
23    // Method to issue the book
24- public void issueBook() {
25        if (!isIssued) {
26            isIssued = true;
27            System.out.println("Book Issued Successfully.");
28        } else {
29            System.out.println("Book is already issued.");
30        }
31    }
32
33    // Method to return the book
34- public void returnBook() {
35        if (isIssued) {
36            isIssued = false;
37            System.out.println("Book Returned Successfully.");
38        } else {
39            System.out.println("Book is already returned.");
40        }
41    }
42
43    // Method to display book details
44- public void displayDetails() {
45        System.out.println("Book Title: " + bookTitle);
46        System.out.println("Author: " + author);
47        System.out.println("Status: " + (isIssued ? "Issued" : "Available"));
48    }
49
50    // Main method
51- public static void main() {
52        LibraryBook lb = new LibraryBook();
53        lb.acceptDetails();
54        lb.issueBook();
55        lb.returnBook();
56        lb.displayDetails();
57    }
58
59 }

```

```

Enter Book Title: JAVA PROGRAMING
Enter Author Name: JAMES GOSLING

----- Book Details -----
Book Title : JAVA PROGRAMING
Author : JAMES GOSLING
Status : Available
Book Issued Successfully.

----- Book Details -----
Book Title : JAVA PROGRAMING
Author : JAMES GOSLING
Status : Issued
Book Returned Successfully.

----- Book Details -----
Book Title : JAVA PROGRAMING
Author : JAMES GOSLING
Status : Available

=== Code Execution Successful ===

```

3. Employee Salary Calculation

```
import java.util.Scanner;
```

```
public class Employee {
```

```
    // Private data members (Data Hiding)
```

```
    private String empName;
```

```
    private int empId;
```

```
    private double basicSalary;
```

```
    // Method to accept employee details
```

```
    public void acceptDetails() {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter Employee Name: ");
```

```
        empName = sc.nextLine();
```

```
        System.out.print("Enter Employee ID: ");
```

```
        empId = sc.nextInt();
```

```
        System.out.print("Enter Basic Salary: ");
```

```
        basicSalary = sc.nextDouble();
```

```
    }
```

```
    // Method to calculate net salary
```

```
    public double calculateSalary() {
```

```
        double hra = basicSalary * 0.20; // 20% HRA
```

```

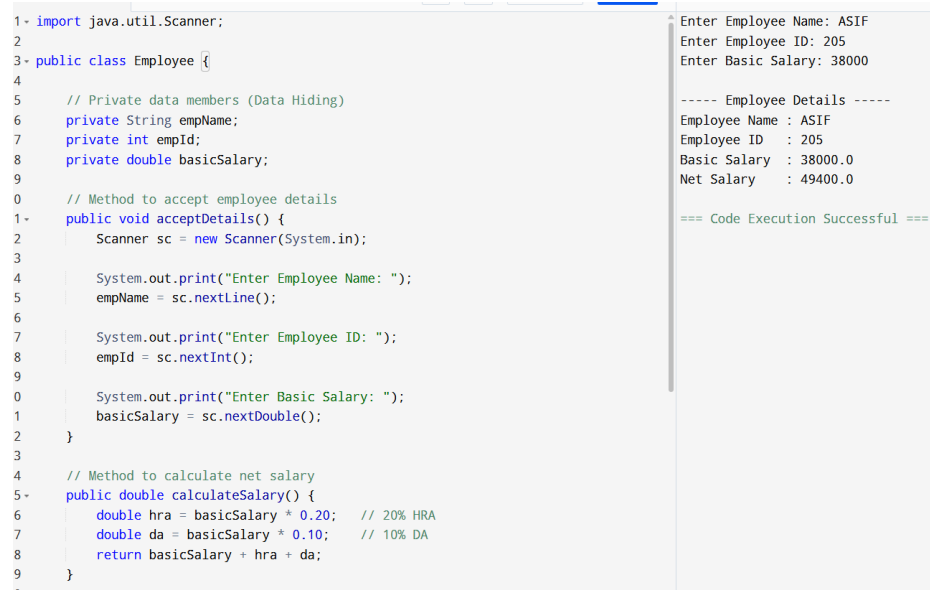
        double da = basicSalary * 0.10; // 10% DA
        return basicSalary + hra + da;
    }

    // Method to display employee details
    public void displayDetails() {
        System.out.println("\n----- Employee Details -----");
        System.out.println("Employee Name : " + empName);
        System.out.println("Employee ID : " + empId);
        System.out.println("Basic Salary : " + basicSalary);
        System.out.println("Net Salary : " + calculateSalary());
    }

    // Main Method
    public static void main(String[] args) {
        Employee e = new Employee();

        e.acceptDetails();
        e.displayDetails();
    }
}

```



The screenshot shows a Java IDE with the following code on the left and its execution output on the right.

```

1- import java.util.Scanner;
2
3- public class Employee {
4
5-     // Private data members (Data Hiding)
6-     private String empName;
7-     private int empId;
8-     private double basicSalary;
9
10-    // Method to accept employee details
11-    public void acceptDetails() {
12-        Scanner sc = new Scanner(System.in);
13
14-        System.out.print("Enter Employee Name: ");
15-        empName = sc.nextLine();
16
17-        System.out.print("Enter Employee ID: ");
18-        empId = sc.nextInt();
19
20-        System.out.print("Enter Basic Salary: ");
21-        basicSalary = sc.nextDouble();
22-    }
23
24-    // Method to calculate net salary
25-    public double calculateSalary() {
26-        double hra = basicSalary * 0.20; // 20% HRA
27-        double da = basicSalary * 0.10; // 10% DA
28-        return basicSalary + hra + da;
29-    }
30- }

```

The output on the right shows the program's execution:

```

Enter Employee Name: ASIF
Enter Employee ID: 205
Enter Basic Salary: 38000

----- Employee Details -----
Employee Name : ASIF
Employee ID : 205
Basic Salary : 38000.0
Net Salary : 49400.0

=== Code Execution Successful ===

```

4. Shape Area Calculator using Polymorphism

```

import java.util.Scanner;

public class Shape {

    public void calculateArea() {
        System.out.println("Area of Shape");
    }
}

```

```
}
```

```
public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    System.out.println("1. Circle");
```

```
    System.out.println("2. Rectangle");
```

```
    System.out.println("3. Triangle");
```

```
    System.out.print("Enter Choice: ");
```

```
    int choice = sc.nextInt();
```

```
    switch(choice) {
```

```
        case 1:
```

```
            System.out.print("Enter Radius: ");
```

```
            double r = sc.nextDouble();
```

```
            double circle = 3.14 * r * r;
```

```
            System.out.println("Area of Circle = " + circle);
```

```
            break;
```

```
        case 2:
```

```
            System.out.print("Enter Length: ");
```

```
            double l = sc.nextDouble();
```

```
            System.out.print("Enter Width: ");
```

```
            double w = sc.nextDouble();
```

```
            double rectangle = l * w;
```

```
            System.out.println("Area of Rectangle = " + rectangle);
```

```
            break;
```

```
        case 3:
```

```
            System.out.print("Enter Base: ");
```

```
            double b = sc.nextDouble();
```

```
            System.out.print("Enter Height: ");
```

```
            double h = sc.nextDouble();
```

```
            double triangle = 0.5 * b * h;
```

```
            System.out.println("Area of Triangle = " + triangle);
```

```
            break;
```

```
        default:
```

```

        System.out.println("Invalid Choice");
    }

    sc.close();
}
}

```



The screenshot shows a Java IDE with two panels. The left panel displays the source code for a `Shape` class. The right panel shows the output of the program execution.

```

1- import java.util.Scanner;
2
3- public class Shape {
4
5-     public void calculateArea() {
6-         System.out.println("Area of Shape");
7-     }
8
9-     public static void main(String[] args) {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.println("1. Circle");
14         System.out.println("2. Rectangle");
15         System.out.println("3. Triangle");
16         System.out.print("Enter Choice: ");
17
18         int choice = sc.nextInt();
19
20         switch(choice) {
21

```

Output:

```

1. Circle
2. Rectangle
3. Triangle
Enter Choice: 1
Enter Radius: 7
Area of Circle = 153.86

=== Code Execution Successful ===

```

5. Vehicle Registration System

```
import java.util.Scanner;
```

```
public class Vehicle {
```

```
    // Private data members
```

```
    private String ownerName;
```

```
    private String vehicleNumber;
```

```
    private String vehicleType;
```

```
    // Method to accept vehicle details
```

```
    public void registerVehicle() {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter Owner Name: ");
```

```
        ownerName = sc.nextLine();
```

```
        System.out.print("Enter Vehicle Number: ");
```

```
        vehicleNumber = sc.nextLine();
```

```
        System.out.print("Enter Vehicle Type: ");
```

```
        vehicleType = sc.nextLine();
```

```
    }
```

```
    // Method to display vehicle details
```

```
    public void displayDetails() {
```

```
        System.out.println("\n----- Vehicle Details -----");
```

```
        System.out.println("Owner Name    : " + ownerName);
```

```
        System.out.println("Vehicle Number : " + vehicleNumber);
```

```
        System.out.println("Vehicle Type  : " + vehicleType);
```

```

    }

    // Main Method
    public static void main(String[] args) {

        Vehicle v = new Vehicle();

        v.registerVehicle();
        v.displayDetails();
    }
}

```



The screenshot shows a Java IDE with two panels. The left panel displays the source code for `Vehicle.java`, and the right panel shows the program's output.

```

1  Vehicle.java java.util.Scanner;
2
3  public class Vehicle {
4
5      // Private data members
6      private String ownerName;
7      private String vehicleNumber;
8      private String vehicleType;
9
10     // Method to accept vehicle details
11     public void registerVehicle() {
12         Scanner sc = new Scanner(System.in);
13
14         System.out.print("Enter Owner Name: ");
15         ownerName = sc.nextLine();
16
17         System.out.print("Enter Vehicle Number: ");
18         vehicleNumber = sc.nextLine();
19
20         System.out.print("Enter Vehicle Type: ");
21         vehicleType = sc.nextLine();
22     }
23

```

The output panel shows the following text:

```

Enter Owner Name: ASIF KHAN
Enter Vehicle Number: TN46AS0805
Enter Vehicle Type: CAR

----- Vehicle Details -----
Owner Name      : ASIF KHAN
Vehicle Number  : TN46AS0805
Vehicle Type    : CAR

=== Code Execution Successful ===

```

6. Bank Account Hierarchy

```

import java.util.Scanner;

// Superclass
class BankAccount {
    protected String accountHolder;
    protected double balance;

    public void acceptDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Account Holder Name: ");
        accountHolder = sc.nextLine();

        System.out.print("Enter Balance: ");
        balance = sc.nextDouble();
    }

    public void displayDetails() {
        System.out.println("\nAccount Holder : " + accountHolder);
        System.out.println("Balance      : " + balance);
    }
}

```

```
// Savings Account
class SavingsAccount extends BankAccount {

    @Override
    public void displayDetails() {
        System.out.println("\nSavings Account");
        System.out.println("Account Holder : " + accountHolder);
        System.out.println("Balance      : " + balance);
    }
}
```

```
// Current Account
class CurrentAccount extends BankAccount {

    @Override
    public void displayDetails() {
        System.out.println("\nCurrent Account");
        System.out.println("Account Holder : " + accountHolder);
        System.out.println("Balance      : " + balance);
    }
}
```

```
// Main Class
public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("1. Savings Account");
        System.out.println("2. Current Account");
        System.out.print("Enter Choice: ");

        int choice = sc.nextInt();
        sc.nextLine();

        BankAccount account;

        if (choice == 1)
            account = new SavingsAccount();
        else
            account = new CurrentAccount();

        account.acceptDetails();
        account.displayDetails();
    }
}
```

}

```
1- import java.util.Scanner;
2
3 // Superclass
4- class BankAccount {
5     protected String accountHolder;
6     protected double balance;
7
8- public void acceptDetails() {
9     Scanner sc = new Scanner(System.in);
10
11     System.out.print("Enter Account Holder Name: ");
12     accountHolder = sc.nextLine();
13
14     System.out.print("Enter Balance: ");
15     balance = sc.nextDouble();
16 }
17
18- public void displayDetails() {
19     System.out.println("\nAccount Holder : " + accountHolder);
20     System.out.println("Balance : " + balance);
21 }
22 }
23
```

```
1. Savings Account
2. Current Account
Enter Choice: 1
Enter Account Holder Name: ASIF
Enter Balance: 100000

Savings Account
Account Holder : ASIF
Balance : 100000.0

=== Code Execution Successful ===
```

7. Hospital Patient Record System

```
import java.util.Scanner;
```

```
public class Patient {
```

```
    // Private data members
```

```
    private String patientName;
```

```
    private int patientId;
```

```
    private int age;
```

```
    private String disease;
```

```
    // Method to accept patient details
```

```
    public void acceptDetails() {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter Patient Name: ");
```

```
        patientName = sc.nextLine();
```

```
        System.out.print("Enter Patient ID: ");
```

```
        patientId = sc.nextInt();
```

```
        System.out.print("Enter Age: ");
```

```
        age = sc.nextInt();
```

```
        sc.nextLine();
```

```
        System.out.print("Enter Disease: ");
```

```
        disease = sc.nextLine();
```

```
    }
```

```
    // Method to display patient details
```

```
    public void displayDetails() {
```

```
        System.out.println("\n----- Patient Record -----");
```

```
        System.out.println("Patient Name : " + patientName);
```

```
        System.out.println("Patient ID : " + patientId);
```

```

        System.out.println("Age      : " + age);
        System.out.println("Disease   : " + disease);
    }

    // Main Method
    public static void main(String[] args) {

        Patient p = new Patient();

        p.acceptDetails();
        p.displayDetails();
    }
}

```



```

1- import java.util.Scanner;
2-
3- public class Patient {
4-
5-     // Private data members
6-     private String patientName;
7-     private int patientId;
8-     private int age;
9-     private String disease;
10-
11-     // Method to accept patient details
12-     public void acceptDetails() {
13-         Scanner sc = new Scanner(System.in);
14-
15-         System.out.print("Enter Patient Name: ");
16-         patientName = sc.nextLine();
17-
18-         System.out.print("Enter Patient ID: ");
19-         patientId = sc.nextInt();
20-
21-         System.out.print("Enter Age: ");
22-         age = sc.nextInt();
23-         sc.nextLine();
24-
25-         System.out.print("Enter Disease: ");
26-         disease = sc.nextLine();
27-     }
}

```

Enter Patient Name: ASIF
Enter Patient ID: 205
Enter Age: 24
Enter Disease: FEVER

----- Patient Record -----
Patient Name : ASIF
Patient ID : 205
Age : 24
Disease : FEVER

=== Code Execution Successful ===

8. E-Commerce Product Catalog

```

import java.util.Scanner;

public class Product {

    // Private data members
    private int productId;
    private String productName;
    private double price;
    private int quantity;

    // Method to accept product details
    public void acceptDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Product ID: ");
        productId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Product Name: ");
    }
}

```

```

        productName = sc.nextLine();

        System.out.print("Enter Product Price: ");
        price = sc.nextDouble();

        System.out.print("Enter Quantity: ");
        quantity = sc.nextInt();
    }

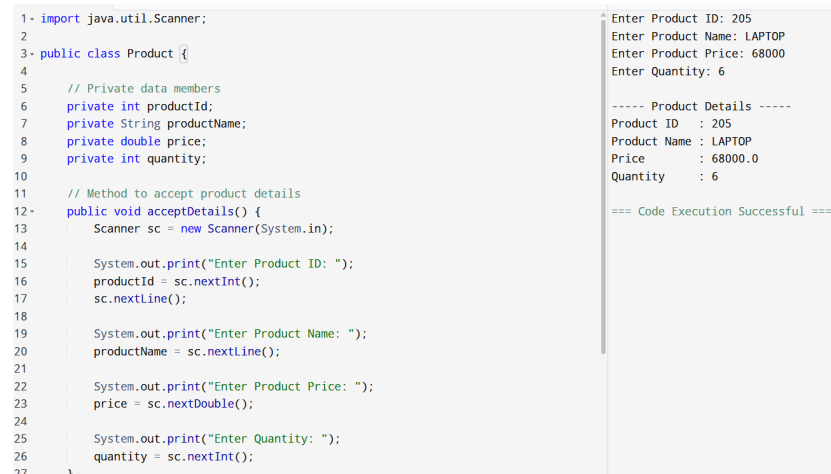
    // Method to display product details
    public void displayDetails() {
        System.out.println("\n----- Product Details -----");
        System.out.println("Product ID   : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Price       : " + price);
        System.out.println("Quantity    : " + quantity);
    }

    // Main Method
    public static void main(String[] args) {

        Product p = new Product();

        p.acceptDetails();
        p.displayDetails();
    }
}

```



The screenshot shows a Java IDE with two panels. The left panel displays the source code for a Java program that uses a Scanner to collect product details (ID, Name, Price, Quantity) and then displays them. The right panel shows the output of the program, which matches the code's logic: it prompts for and displays the entered values for a product named 'LAPTOP' with ID 205, price 68000, and quantity 6. The output ends with '=== Code Execution Successful ==='.

```

1- import java.util.Scanner;
2
3- public class Product {
4
5     // Private data members
6     private int productId;
7     private String productName;
8     private double price;
9     private int quantity;
10
11     // Method to accept product details
12     public void acceptDetails() {
13         Scanner sc = new Scanner(System.in);
14
15         System.out.print("Enter Product ID: ");
16         productId = sc.nextInt();
17         sc.nextLine();
18
19         System.out.print("Enter Product Name: ");
20         productName = sc.nextLine();
21
22         System.out.print("Enter Product Price: ");
23         price = sc.nextDouble();
24
25         System.out.print("Enter Quantity: ");
26         quantity = sc.nextInt();
27     }
}

```

```

Enter Product ID: 205
Enter Product Name: LAPTOP
Enter Product Price: 68000
Enter Quantity: 6

----- Product Details -----
Product ID   : 205
Product Name : LAPTOP
Price       : 68000.0
Quantity    : 6

=== Code Execution Successful ===

```

9. University Staff Hierarchy

```

import java.util.Scanner;

// Superclass
class Staff {
    protected String name;

```

```

protected int id;

public void acceptDetails() {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Staff Name: ");
    name = sc.nextLine();

    System.out.print("Enter Staff ID: ");
    id = sc.nextInt();
}

public void displayDetails() {
    System.out.println("Name : " + name);
    System.out.println("ID : " + id);
}
}

// Subclass: Professor
class Professor extends Staff {

    @Override
    public void displayDetails() {
        System.out.println("\nProfessor Details");
        super.displayDetails();
    }
}

// Subclass: Lecturer
class Lecturer extends Staff {

    @Override
    public void displayDetails() {
        System.out.println("\nLecturer Details");
        super.displayDetails();
    }
}

// Main Class
public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

```

```

        System.out.println("1. Professor");
        System.out.println("2. Lecturer");
        System.out.print("Enter Choice: ");

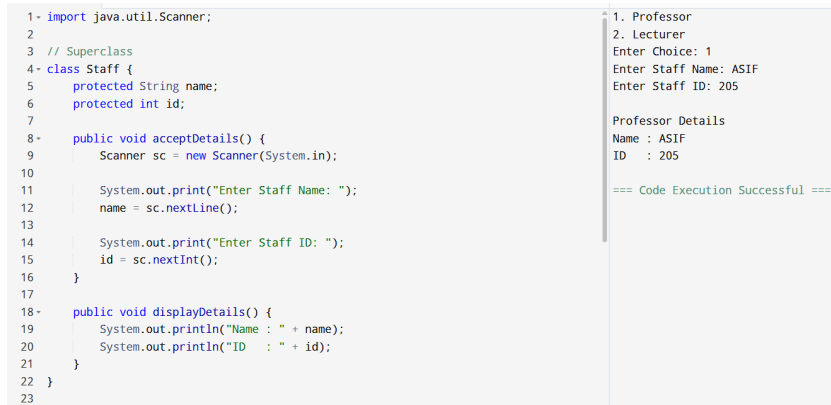
        int choice = sc.nextInt();
        sc.nextLine();

        Staff staff;

        if (choice == 1)
            staff = new Professor();
        else
            staff = new Lecturer();

        staff.acceptDetails();
        staff.displayDetails();
    }
}

```



```

1- import java.util.Scanner;
2
3 // Superclass
4- class Staff {
5     protected String name;
6     protected int id;
7
8- public void acceptDetails() {
9     Scanner sc = new Scanner(System.in);
10
11     System.out.print("Enter Staff Name: ");
12     name = sc.nextLine();
13
14     System.out.print("Enter Staff ID: ");
15     id = sc.nextInt();
16 }
17
18- public void displayDetails() {
19     System.out.println("Name : " + name);
20     System.out.println("ID : " + id);
21 }
22 }
23
1. Professor
2. Lecturer
Enter Choice: 1
Enter Staff Name: ASIF
Enter Staff ID: 205

Professor Details
Name : ASIF
ID : 205

=== Code Execution Successful ===

```

10. Smart Home Device Controller

```

import java.util.Scanner;

// Superclass
class SmartDevice {

    public void executeCommand(String cmd) {
        System.out.println("Unknown Command");
    }
}

// Smart Light
class SmartLight extends SmartDevice {

    @Override
    public void executeCommand(String cmd) {

```

```

        if (cmd.equalsIgnoreCase("on"))
            System.out.println("Smart Light is ON");
        else if (cmd.equalsIgnoreCase("off"))
            System.out.println("Smart Light is OFF");
        else if (cmd.equalsIgnoreCase("status"))
            System.out.println("Smart Light Status: Working");
        else
            System.out.println("Invalid Command");
    }
}

```

// Smart Thermostat

```

class SmartThermostat extends SmartDevice {

```

```

    @Override
    public void executeCommand(String cmd) {

        if (cmd.equalsIgnoreCase("on"))
            System.out.println("Thermostat is ON");
        else if (cmd.equalsIgnoreCase("off"))
            System.out.println("Thermostat is OFF");
        else if (cmd.equalsIgnoreCase("status"))
            System.out.println("Thermostat Status: 24°C");
        else
            System.out.println("Invalid Command");
    }
}

```

// Smart Lock

```

class SmartLock extends SmartDevice {

```

```

    @Override
    public void executeCommand(String cmd) {

        if (cmd.equalsIgnoreCase("on"))
            System.out.println("Smart Lock Locked");
        else if (cmd.equalsIgnoreCase("off"))
            System.out.println("Smart Lock Unlocked");
        else if (cmd.equalsIgnoreCase("status"))
            System.out.println("Smart Lock Status: Secure");
        else
            System.out.println("Invalid Command");
    }
}

```

```

// Main Class
public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Choose Device");
        System.out.println("1. Smart Light");
        System.out.println("2. Smart Thermostat");
        System.out.println("3. Smart Lock");
        System.out.print("Enter Choice: ");

        int choice = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Command (on/off/status): ");
        String cmd = sc.nextLine();

        SmartDevice device;

        switch (choice) {
            case 1:
                device = new SmartLight();
                break;

            case 2:
                device = new SmartThermostat();
                break;

            case 3:
                device = new SmartLock();
                break;

            default:
                System.out.println("Invalid Choice");
                return;
        }

        // Polymorphism
        device.executeCommand(cmd);
    }
}

```

```

1- import java.util.Scanner;
2
3 // Superclass
4- class SmartDevice {
5
6-     public void executeCommand(String cmd) {
7         System.out.println("Unknown Command");
8     }
9 }
10
11 // Smart Light
12- class SmartLight extends SmartDevice {
13
14     @Override
15-     public void executeCommand(String cmd) {
16
17         if (cmd.equalsIgnoreCase("on"))
18             System.out.println("Smart Light is ON");
19         else if (cmd.equalsIgnoreCase("off"))
20             System.out.println("Smart Light is OFF");
21         else if (cmd.equalsIgnoreCase("status"))
22             System.out.println("Smart Light Status: Working");
23         else
24             System.out.println("Invalid Command");
25     }
26 }

```

```

Choose Device
1. Smart Light
2. Smart Thermostat
3. Smart Lock
Enter Choice: 1
Enter Command (on/off/status): on
Smart Light is ON

=== Code Execution Successful ===

```